

Joseph Simeon

Electronics engineer, Brisbane, QLD.

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Electronics engineer with over 4 years of expertise in developing firmware and software for embedded systems; programming in C & C++ in application and drivers in both bare-metal & RTOS which involved testing and supporting a smart touch LCD for a lighting control company using STM32 hardware and ST's graphical framework TouchGFX. A strong background in electronics with a focus on firmware and software programming, debugging and release testing a product that also involved supplying customer support for the products usage as well as fixing issues.

Skills

- Programming in C/C++ for embedded systems (bare-metal, FreeRTOS) with knowledge in Unix, Python and Rust.
 - Supplemental knowledge in HTML/CSS/JS and SQL.
- Industry experience in application and driver programming for peripherals such as UART, I2C, SPI, QPSI, DSI, DMA, EEPROM, LCDs (MipiDSI), PWM, Timers, ADC, GPIO.
- Debugging and testing within industry hardware and software such as Wireshark, logic analysers, TTL, oscilloscopes, JTAG, ST-LINK, Ozone, C-Lion using GDB.
- Control experience with docker, git, Jenkins utilising ticketing and tracking with Gitlab and Redmine.
- Writing documentation with project development in issue tracking, factory and site testing, release testing, and manuals.
- Creating an embedded graphical user interface with custom widget/events within graphical framework.
- Providing customer and integrator support related to a product.

Experience

Firmware Engineer, Zencontrol, Brisbane

October 2020 — December 2024

- Handled the development for smart touch LCD project, creating UI/UX with custom widgets and event functionality using TouchGX graphical framework that interacted with a DALI controlled lighting systems on STM32 hardware with LCD.
- Programming, debugging, and release testing firmware built in C and C++ on STM32 hardware.
 - Testing the product in factory and sites settings.
- Developing driver code for the product, as well as viewing schematics and supplying reports for part validation.
- Documentation writing on project development, wikipedia style manuals, and testing tickets.
- Providing custom support, collaborating with clients in troubleshooting issues to be turned into fixes.

Firmware Engineer (Thesis), Griffith University, Brisbane

January 2020 — July 2020

- Creating a digital instrument cluster using Raspberry Pi and LCD with a graphical interface using JavaFX.
- Up-skilling in all areas needed to complete the project such as learning a new language and framework.
- Independent research, utilising information to prove design decisions.
- Reading documentation such as data-sheets, standards, and equipment manuals.

Education

Bachelor of Engineering with Honours in Electronics and Energy Engineering

Griffith University

2016 — 2021

- Assisted in creation of the SCADA course for Griffith University with other selected students.
- Membership of the IEEE society at Griffith University as a 2nd year — 4th year student, assisted in open events and member activities within Griffith University campus.

Diploma of Engineering

Griffith College (formerly QIBT)

2014 — 2015

References at request.